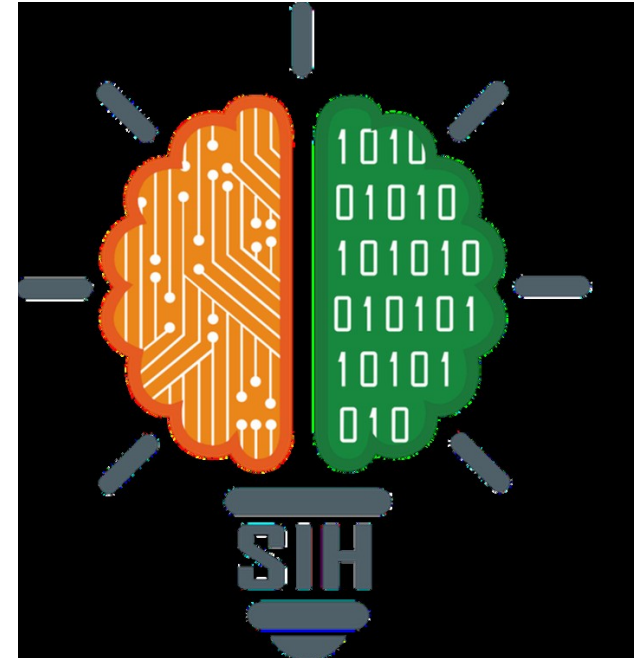


SMART INDIA HACKATHON 2026



- **PROBLEM STATEMENT ID** - 26126
- **PROBLEM STATEMENT TITLE** - VISION BASED AUTONOMOUS NAVIGATION FOR UNMANNED GROUND VEHICLE FOR OUTDOOR ENVIRONMENT
- **THEME** - Smart Automation
- **PS Category** - Software
- **Team ID** - 137858
- **TEAM NAME** - HACKTIDE



See • Assess • Adapt — camera-only navigation that knows how much to trust its own eyes

One closed loop: perceive terrain, score risk and uncertainty, localise without GPS, replan on change, reach B safely.

END-TO-END AUTONOMY LOOP



Perceive

Lightweight segmentation labels grass, dirt, gravel, mud, rock, tree, ditch.



Assess

Energy score flags unknown pixels; a learned regressor rates frame confidence.



Localise

Visual odometry / SLAM + wheel odometry; tracking health monitored.



Map

Dynamic risk map 0–100 per cell: terrain, geometry, obstacle, unknown, staleness.



Plan

Global A* on risk cost fires on events; local arc planner at 10 Hz.



Move

Governor sets speed and margins from confidence; controller drives the UGV.

EXISTING APPROACH VS DRISHTI-NAV

Existing Approach

- ✗ Binary free / blocked map
- ✗ Shortest path at full speed
- ✗ 'Not detected' treated as safe
- ✗ Blind to lighting changes
- ✗ Full replan every frame
- ✗ No measure of self-trust



DRISHTI-Nav

- ✓ Continuous risk field (0–100)
- ✓ Safer route, not just shorter
- ✓ Unknown → uncertain → avoid
- ✓ Confidence-aware speed & margin
- ✓ Event-driven replanning
- ✓ Localisation health → recovery

INNOVATION & UNIQUENESS



Multi-layer dynamic risk map

Terrain + geometry + obstacle + unknown fused per cell, updated every frame.



Unknown-aware perception

Energy-score OOD + 'novel' class: unfamiliar objects get risk, not a free pass.



Perception-confidence governor

Learned frame-quality score drives speed, safety margin and risk weight.



Event-driven replanning

Global A* only when the path corridor changes; local arcs at 10 Hz.



Lighting-adaptive behaviour

Shadows / low light → lower confidence → conservative navigation.

SAFETY BY DESIGN



Fail-safe autonomy

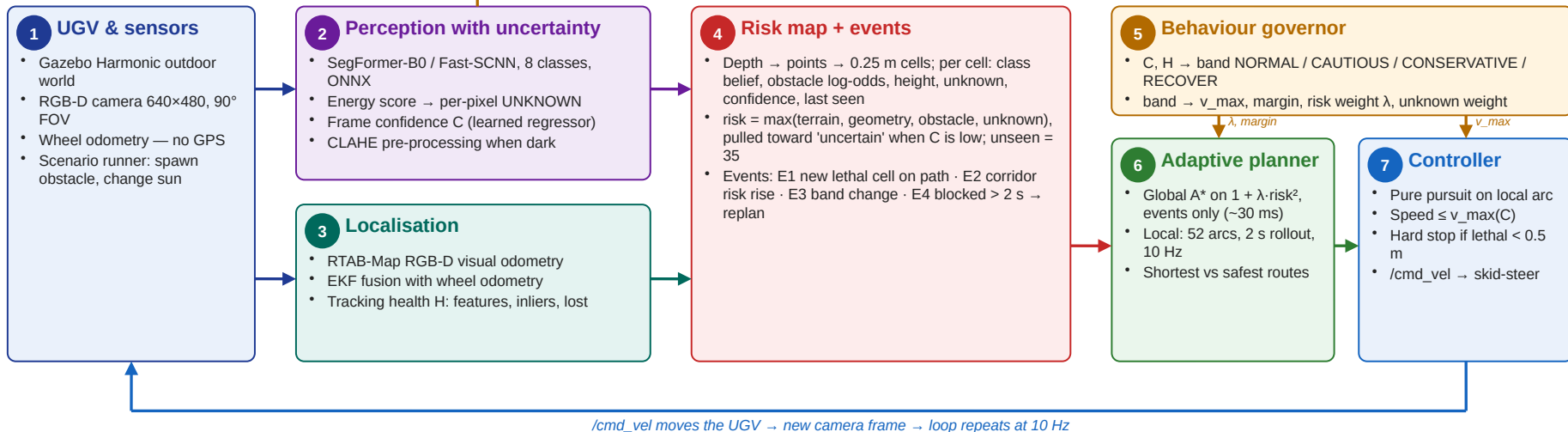
Every uncertainty has a response

- ✓ Unseen cells carry risk, never 'free'
- ✓ Hard stop on lethal cell ahead
- ✓ Low confidence → slow + wide margin
- ✓ SLAM lost → hold, re-acquire
- ✓ Evidence accumulates, no phantom stops
- ✓ Works with RGB-D / stereo, no GPS



FLOW AND ARCHITECTURE

confidence C (from 2) + localisation health H (from 3)



WHAT HAPPENS WHEN THE WORLD CHANGES



Rock appears 7 m ahead

detected in 1 frame → cell risk 92 → E1
→ A^* replan ≈ 30 ms → UGV passes with margin



Unfamiliar object

no class fits → energy score high → UNKNOWN → risk 60 + 50· u → treated as lethal, wider berth



Sun dims / shadows

confidence 0.95 → 0.47 → CONSERVATIVE: v 1.6 → 0.5 m/s, margin ×2, only risk < 66 driven



Light returns

confidence recovers → NORMAL band → speed resumes → Point B reached, zero collisions



PROTOTYPE VIDEO

Closed-loop simulation: risk map, unknown object, lighting event, replanning, $A \rightarrow B$.

[YouTube link – add before submission]



TECHNOLOGIES USED



Simulation

ROS 2 Jazzy · Gazebo Harmonic · segmentation & RGB-D camera sensors · scenario runner (YAML)



Perception AI

PyTorch · SegFormer-B0 (3.7 M params) / Fast-SCNN · energy-score OOD · ONNX Runtime / TensorRT



Localisation

RTAB-Map (RGB-D) · robot_localization EKF · own ORB + PnP visual odometry as fallback



Planning & control

NumPy multi-layer risk map · A* (scikit-image / pyastar2d) · arc-sampling local planner · pure pursuit



Dashboard & eval

Foxglove Studio · FastAPI + WebSocket web view · seeded evaluation harness · CSV metrics



Proof-of-concept (now)

Pure Python: NumPy · SciPy · scikit-image · Matplotlib renderer · imageio (demo video, slide 6)

WHY THIS IS BUILDABLE BY A STUDENT TEAM

- Simulator auto-labels every training frame → no manual annotation; RUGD / RELIS-3D / ORFD for real-image fine-tuning.
- Every module is NumPy-level maths on a 0.25 m grid; A* replans in ~30 ms, whole loop runs at 10 Hz on a laptop CPU.
- Same ROS 2 nodes run on a Jetson with a RealSense / ZED camera — the simulator is swapped, not the code.
- Working proof-of-concept already exists (video, slide 6): 45 seeded runs, 3 planner variants.

POTENTIAL CHALLENGES & RISKS

- Textureless grass and sky make visual SLAM lose tracking outdoors.
- Segmentation degrades under strong shadows, dusk and glare.
- Objects never seen in training are misclassified as 'safe'.
- Sim-to-real gap: simulator textures differ from real terrain.

ANALYSIS OF FEASIBILITY

- All components are proven open-source parts (RTAB-Map, ROS 2, SegFormer) — the novelty is in how they are combined.
- Risk table, bands and thresholds live in YAML — tunable without retraining.
- Prototype loop runs 10 Hz on CPU; model is ~3.7 M params → embedded-ready (ONNX / TensorRT).
- 10-week plan: A → B with ground-truth pose by week 3, full stack by week 8.

STRATEGIES FOR OVERCOMING RISKS

- Fuse VO with wheel odometry (EKF); tracking-health signal → hold and re-acquire.
- Confidence governor slows and widens margins when lighting degrades; CLAHE pre-processing.
- Energy-score OOD + 'novel' class trained by outlier exposure in simulation → unknown gets risk.
- Domain randomisation (sun, textures, fog) + fine-tune on RUGD / RELIS-3D real images.

IMPACT AND BENEFITS



IMPACT AREAS — what changes for outdoor UGV operations



GPS-denied defence mobility

UGVs keep moving under jamming, in forests, valleys and urban canyons where GNSS fails — BEL's core use case.



Search & rescue

Reaches casualties over rubble and unstable ground; unknown debris is avoided, not driven into.



Agriculture

Field robots choose firm ground over mud, cutting bogging and crop damage; runs on a low-cost camera.



Delivery, surveillance, mining

Perimeter patrol and last-mile logistics at dawn, dusk and under tree cover with adaptive caution.

KEY PROGRAM IMPACTS

Collision-free autonomy

0 collisions in 45 seeded runs; unknown objects and ditches avoided, not assumed safe.

Safer, not just shorter

Time in mud cut from ~40 s to under 1 s per run vs. binary planner; path ~13 % longer.

Self-aware perception

Confidence score tells the vehicle when to slow down — a first-class signal.

Fast reaction

New hazard detected within 1–2 frames, ~100–200 ms (binary map: ~960 ms); A* replan in under 30 ms.

Low cost, lightweight

One RGB-D camera, no LiDAR, no GPS; 3.7 M-param model runs on Jetson-class boards.

Measurable & reproducible

Seeded scenarios and ablations; every claim is a number from the harness.

TARGET AUDIENCE

1

BEL / defence UGV programmes

Camera-first autonomy module for GPS-denied outdoor platforms.

2

Disaster-response agencies (NDRF, SDRF)

Ground robots for reconnaissance over unstable terrain.

3

Agri-robotics & plantation operators

Low-cost navigation for field and orchard vehicles.

4

Logistics, mining & surveillance

Perimeter and last-mile autonomy in unstructured sites.

5

Robotics researchers & students

Open, simulator-agnostic reference stack with evaluation harness.

WHO GAINS WHAT

Operator

Vehicle reaches the objective without babysitting; risk appetite is one slider.

Platform maker

Drop-in software module; camera only, no GPS/LiDAR bill of materials.

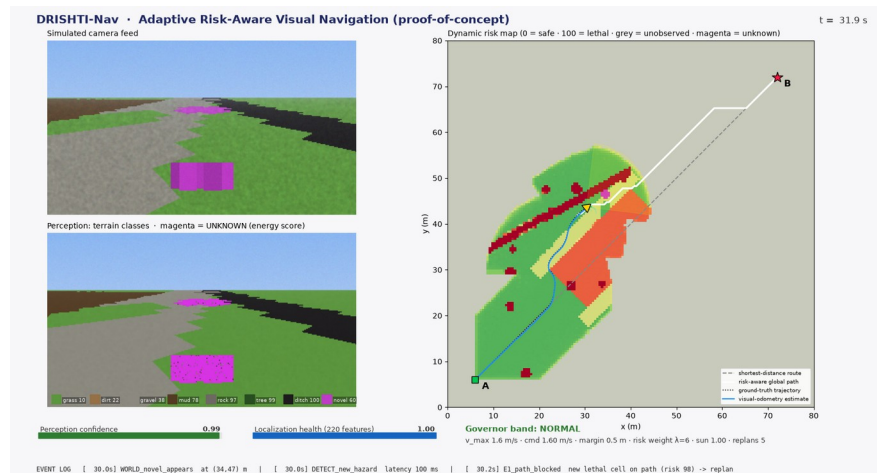
Mission

Fewer stuck or damaged vehicles; predictable, explainable behaviour.

Right terrain · right speed · right margin — collision-free Point A → Point B without GPS



PROJECT LINKS & DEMO



Proof-of-concept dashboard (t = 32 s): an unfamiliar object is flagged UNKNOWN (magenta) in the camera and the map; the risk-aware route (white) leaves the mud belt (orange) that the shortest route (grey) crosses; VO estimate vs ground truth, confidence gauge, governor band and event log.

PROTOTYPE RESULTS — 15 seeded runs per planner, same world

Metric	Binary planner	Risk-aware	DRISHTI-Nav
Success (no collision)	100 %	100 %	100 %
Runs with collision	0/15	0/15	0/15
Time driving in mud (s, mean)	39.7 s	1.6 s	0.6 s
Hazard detection latency (ms)	960	100	213
Global replans / run	43.1	5.3	23.2
Path length (m)	94.1	101.8	106.2

PROTOTYPE VIDEO



Full walkthrough of the working simulation (4 min) · [YouTube link – add before submission]

GIT-HUB LINK



Complete repository — simulator, risk map, planner, evaluation harness, deck generator

https://github.com/kiransandilla/SIH_PS_126

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